



DEPARTMENT OF BIOMEDICAL ENGINEERING
QUESTION BANK

Regulation	:	2023
Course Name	:	Food & Nutrition
Course Code	:	JBM9201
Year / Semester / Branch	:	III/IV/AI & ML , CS and CSE
Unit No	:	1
Unit Title	:	Food, Nutrition and Health
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define the term food. State its importance. Food is any substance consumed to provide nutrients and energy. It supports growth, repair, and maintenance of the body. Food supplies carbohydrates, proteins, fats, vitamins, and minerals. It helps regulate body processes. Adequate food intake ensures good health.	2	K1	CO1
2	Define nutrition? Nutrition is the science of food and its relationship to health. It studies nutrients and their utilization in the body. Proper nutrition maintains growth and development. It prevents deficiency diseases. Nutrition is essential for overall well-being.	2	K1	CO1
3	Explain the relationship between food and health. Food directly influences physical and mental health. Balanced diet prevents malnutrition and diseases. Poor food habits lead to deficiency disorders. Good nutrition improves immunity. Thus, food is vital for maintaining health.	2	K2	CO1
4	Define energy and specify its role in the body. Energy is the capacity to do work. It is required for body movements and metabolism. Energy maintains body temperature. It supports growth and repair. Energy comes mainly from carbohydrates, fats, and proteins.	2	K1	CO1

5	Mention the units of energy measurement. Energy is measured in kilocalories (kcal). Another unit is kilojoule (kJ). 1 kcal equals 4.184 kJ. These units measure energy from food. They indicate fuel value of nutrients.	2	K1	CO1
6	List the various forms of energy in the body present in the human body. Chemical energy from food is primary. Mechanical energy enables movement. Thermal energy maintains body temperature. Electrical energy supports nerve impulses. All forms are essential for survival.	2	K1	CO1
7	Explain the role of carbohydrates in energy requirements of a physiological system. Carbohydrates are primary fuel. They provide quick energy. Stored as glycogen. Prevent protein breakdown. Essential for brain function.	2	K2	CO1
8	Define the term health. Health is a state of well-being. Includes physical, mental, and social aspects. Not merely absence of disease. Depends on nutrition and lifestyle. Essential for quality life.	2	K1	CO1
9	Why is balanced diet important? Provides all nutrients in right amount. Maintains energy balance. Prevents deficiency diseases. Supports growth and immunity. Promotes overall health.	2	K2	CO1
10	State how energy value of foods are determined? The energy value of foods is determined by the calorimetry method. In this method, a known quantity of food is completely burnt in a bomb calorimeter. The heat released during combustion is measured. This heat indicates the gross energy content of the food. The value is expressed in kilocalories or kilojoules.	2	K2	CO1
11	List the various sources of dietary energy. Carbohydrates are main source. Fats provide concentrated energy. Proteins provide limited energy. Alcohol also provides energy. Balanced intake is essential.	2	K1	CO1
12	Define the term diet-induced thermogenesis. It is energy spent in digestion of food. Includes absorption and metabolism. Also called thermic effect of food. Increases metabolic rate	2	K1	CO1

	after meals. Essential part of energy expenditure.			
13	Indicate the factors that affect energy requirements. Age influences energy needs. Physical activity increases energy demand. Body size and sex matter. Climate affects energy expenditure.	2	K1	CO1
14	Define physiological fuel value. It is the energy released from nutrients during metabolism. Carbohydrates provide 4 kcal/g. Proteins provide 4 kcal/g. Fats provide 9 kcal/g. It indicates usable energy for the body.	2	K1	CO1
15	Why is energy important for health? Energy supports body functions. It enables physical activity. Maintains body temperature. Supports growth and repair. Energy deficiency causes fatigue.	2	K2	CO1
16	Define energy balance. Energy balance is intake vs expenditure. Positive balance causes weight gain. Negative balance causes weight loss. Balance maintains body weight. Important for health management.	2	K1	CO1
17	List the factors that affect BMR. Age affects metabolic rate. Body size and composition influence BMR. Sex affects energy needs. Hormonal activity plays a role. Climate and illness also affect BMR.	2	K1	CO1
18	Define Basal Metabolic Rate (BMR). BMR is the minimum energy required at rest. It maintains vital body functions. Functions include breathing and circulation. Measured under standard conditions. BMR varies with age and body size.	2	K1	CO1
19	Explain physiological energy. Physiological energy is usable energy. Available after digestion and absorption. Depends on nutrient metabolism Differs from gross energy. Important for dietary planning.	2	K2	CO1
20	A Patient's weight is 70 kg and height is 1.70 m. Calculate BMI. BMI = Weight (kg) / [Height (m)] ² - Metric Units BMI = 70 / (1.70 × 1.70) = 24.22 kg/m ²	2	K2	CO1

PART B

Q.No.	Questions	Marks	BTL	CO
1	Interpret the relationship between food, nutrition, and health in detail.	16	K2	CO1
2	Explain the various form of energy in the human body. Discuss the units of energy measurement and physiological fuel values of nutrients.	16	K2	CO1
3	Describe how energy value of foods are determined using calorimetry. Add a note on gross and physiological energy values.	16	K2	CO1
4	Define Basal Metabolic Rate (BMR). Explain the methods of determination and the factors affecting BMR.	16	K2	CO1
5	Summarize the factors affecting energy requirements of individuals. Explain diet-induced thermogenesis in detail.	16	K2	CO1
6	Explain the concept of energy balance. Discuss the effects of positive and negative energy balance on health.	16	K2	CO1

COURSE FACULTY

COURSE COORDINATOR

HOD

SCRUTINIZER



DEPARTMENT OF BIOMEDICAL ENGINEERING
QUESTION BANK

Regulation	:	R2023
Course Name	:	FOOD & NUTRITION
Course Code	:	JBM9201
Year / Semester / Branch	:	III/VI/ AI & ML, CS and CSE
Unit No	:	2
Unit Title	:	MACRONUTRIENTS
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define macronutrients. Macronutrients are nutrients required in large amounts by the body. They provide energy and structural components. Main macronutrients are carbohydrates, proteins, and fats. They are essential for growth, repair, and metabolic activities. Adequate intake maintains overall health.	2	K1	CO2
2	List the macronutrients. Macronutrients include carbohydrates, proteins, and fats. Carbohydrates are the primary energy source. Proteins are essential for body building. Fats provide concentrated energy. All three are required for balanced nutrition.	2	K1	CO2
3	State one function of carbohydrates. Carbohydrates provide energy for daily activities. They are stored as glycogen in liver and muscles. Carbohydrates spare proteins from energy use. They are vital for brain function. Adequate intake prevents fatigue.	2	K1	CO2
4	List two sources of carbohydrates. Sources of carbohydrates include rice and wheat. Other sources are bread and potatoes. Fruits like banana are rich in carbs. Legumes also contain carbohydrates. They provide energy and dietary fiber, supporting digestion and metabolism. Including a variety of carbohydrate sources ensures a steady supply of energy for body functions.	2	K1	CO2
5	Define protein. Proteins are made of amino acids and build body tissues. They support growth and repair. Proteins are also involved in enzyme and hormone	2	K1	CO2

	synthesis. They provide limited energy. Proteins are essential for overall health.			
6	State one function of protein. Proteins help in muscle development and tissue repair. They are needed for enzyme and hormone formation. Proteins maintain body structure. They support immunity. Adequate protein intake is essential for growth.	2	K1	CO2
7	List two protein-rich foods. Pulses like lentils are rich in protein. Eggs are another good source. Milk and dairy also provide proteins. Nuts and seeds contain proteins. Meat and fish are animal protein sources.	2	K1	CO2
8	Define fat? Fat is a concentrated source of energy. It provides essential fatty acids. Fat supports insulation and organ protection. Fat-soluble vitamins require fat for absorption. Adequate fat intake is necessary for health.	2	K1	CO2
9	State one function of fat. Fat stores energy for later use. It insulates the body against heat loss. Fat cushions organs. It aids in absorption of vitamins A, D, E, K. Fat is essential for cell membrane structure.	2	K1	CO2
10	Name two fat sources. Butter and cooking oils are rich in fat. Nuts and seeds contain healthy fats. Fatty fish provides omega-3 fatty acids. Dairy products are another source. Meat also contains fats.	2	K1	CO2
11	State the function RDA. RDA stands for Recommended Dietary Allowance. It is the average daily intake sufficient for most healthy individuals. RDA varies with age, sex, and activity. It guides diet planning. Ensures nutrient adequacy.	2	K1	CO2
12	Mention one protein deficiency disease. Kwashiorkor is caused by severe protein deficiency. It is common in children. Symptoms include edema and stunted growth. Hair changes and fatty liver are observed. Adequate protein prevents this disease.	2	K1	CO2
13	Mention one carbohydrate deficiency effect. Carbohydrate deficiency causes fatigue. It leads to weakness and poor concentration. Muscle glycogen depletion occurs. Hypoglycemia may result. Adequate carbs prevent energy loss.	2	K1	CO2
14	Mention one fat deficiency symptom. Fat deficiency may cause dry skin. Hair becomes brittle. Poor vitamin absorption occurs. Weight loss is common. Fat intake is necessary for health.	2	K1	CO2

15	Define dietary fiber? Dietary fiber is indigestible carbohydrate. It improves digestion and bowel movement. Fiber prevents constipation. It helps in cholesterol management. Found in cereals, fruits, and vegetables.	2	K1	CO2
16	List out any two types of dietary fiber Soluble fiber dissolves in water and forms gel. Found in oats and legumes. Insoluble fiber does not dissolve, adds bulk. Found in wheat bran and vegetables. Both improve digestive health.	2	K1	CO2
17	State one function of dietary fiber. Dietary fiber prevents constipation. It aids in maintaining healthy gut microbiota. Helps in blood sugar control. Reduces cholesterol levels. Supports overall digestive health.	2	K1	CO2
18	Define digestion? Digestion is the breakdown of food into absorbable molecules. It occurs in the mouth, stomach, and intestines. Enzymes play a key role. Digestion releases nutrients and energy. It is essential for nutrient absorption.	2	K1	CO2
19	Define absorption? Absorption is uptake of nutrients into the bloodstream. Occurs mainly in the small intestine. Nutrients travel via blood or lymph. Ensures energy and body functions. Essential for maintaining health.	2	K1	CO2
20	Explain why a balanced intake of macronutrients is important. Balanced macronutrients provide energy and body-building materials. Prevent deficiency and excess problems. Supports growth and repair. Maintains immunity and metabolism. Ensures overall health.	2	K2	CO2

PART B

Q. No.	Questions	Marks	BTL	CO
1	Apply the knowledge of macronutrients to analyze their functions, dietary sources, and importance in human health.	16	K3	CO2
2	Analyze the classification of carbohydrates and apply this knowledge to explain their functions, sources, digestion, absorption, Recommended Dietary Allowances (RDA), and deficiency disorders.	16	K4	CO2
3	Apply the concepts of digestion and absorption of proteins and analyze their functions, deficiency conditions, and RDA.	16	K3	CO2
4	Analyze fats in the diet and apply the knowledge to explain their types, functions, dietary sources, and deficiency symptoms.	16	K4	CO2
5	Apply the classification of dietary fiber and analyze its functions, sources, and role in disease prevention.	16	K3	CO2
6	Apply the concept of energy value of macronutrients and analyze energy balance, including positive and negative balance, and their effects on health.	16	K3	CO2
7	Compare carbohydrates, proteins, and fats based on classification, functions, digestion, absorption, energy value, and Recommended Dietary Allowances (RDA).	16	K4	CO2
8	Analyze the deficiency disorders associated with carbohydrates, proteins, and fats and suggest preventive dietary measures.	16	K4	CO2



DEPARTMENT OF BIOMEDICAL ENGINEERING

QUESTION BANK

Regulation	:	2023
Course Name	:	Food & Nutrition
Course Code	:	JBM9201
Year / Semester / Branch	:	II/IV/ AI & ML, CS and CSE
Unit No	:	3
Unit Title	:	Micronutrients- Vitamins and Minerals
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define fat-soluble vitamins. Fat-soluble vitamins are vitamins that dissolve in fat and are absorbed along with dietary lipids. These vitamins are stored in the liver and fatty tissues of the body for long durations. The major fat-soluble vitamins include Vitamin A, D, E, and K. Since they remain stored, excess consumption may lead to toxicity. They play essential roles such as vision maintenance, bone growth, antioxidant activity, and blood clotting. Proper intake is necessary to prevent deficiency and avoid harmful accumulation in the body.	2	K1	CO3
2	List any four fat-soluble vitamins. The four major fat-soluble vitamins are Vitamin A, Vitamin D, Vitamin E, and Vitamin K. These vitamins require the presence of fat in the diet for proper absorption in the intestine. They are stored in the liver and adipose tissue, which makes them available for use when needed. Vitamin A supports vision and immunity, Vitamin D helps calcium absorption and bone health, Vitamin E protects cells as an antioxidant, and Vitamin K plays a key role in blood clotting mechanisms.	2	K1	CO3

3	State the full form of RDA. RDA stands for Recommended Dietary Allowance. It refers to the average daily intake of a nutrient that is sufficient to meet the nutritional needs of nearly all healthy individuals in a specific age group and gender category. RDA values are established based on scientific studies and are used to guide diet planning and nutritional assessment. They help in preventing deficiencies and maintaining proper health. RDAs may differ depending on age, pregnancy, lactation, and physical activity levels.	2	K1	CO3
4	Mention any two water-soluble vitamins. Two examples of water-soluble vitamins are Vitamin C and Vitamin B1 (Thiamin). Water-soluble vitamins dissolve in water and are easily absorbed into the bloodstream. Unlike fat-soluble vitamins, they are not stored in large quantities in the body, and excess amounts are usually excreted through urine. Therefore, regular dietary intake is important. These vitamins play major roles in energy production, tissue repair, immunity, and nerve function. Deficiency may occur quickly if dietary intake is inadequate.	2	K1	CO3
5	Identify the vitamin known as Thiamin. Thiamin is known as Vitamin B1. It is a water-soluble vitamin that plays an essential role in carbohydrate metabolism by helping convert glucose into usable energy. Thiamin is also important for proper nerve function and muscle activity. Deficiency of Vitamin B1 can lead to beriberi, which affects the heart and nervous system, and also Wernicke-Korsakoff syndrome in chronic alcoholism. Major sources of thiamin include whole grains, legumes, nuts, pork, and fortified cereals.	2	K1	CO3
6	Name the vitamin deficiency that causes scurvy. Scurvy is caused by the deficiency of Vitamin C (ascorbic acid). Vitamin C is essential for collagen formation, which is important for healthy gums, skin, and connective tissues. When Vitamin C is deficient, symptoms such as bleeding gums, loosening of teeth, fatigue, weakness, delayed wound healing, and joint pain may occur. Vitamin C also acts as an antioxidant and enhances the absorption of iron in the body. Rich sources include citrus fruits, guava, tomatoes, and green leafy vegetables.	2	K1	CO3
7	List any two macro minerals. Two examples of macro minerals are Calcium and Phosphorus. Macro minerals are required by the body in relatively large amounts compared to trace minerals. Calcium is essential for bone and teeth development, blood clotting, nerve transmission, and muscle contraction. Phosphorus	2	K1	CO3

	supports bone formation, energy production through ATP, and cell membrane structure. Deficiency of these minerals may cause weak bones, muscle weakness, and metabolic disturbances. Dairy products, meat, legumes, nuts, and fish are major sources of these minerals.			
8	Name any two micro minerals (trace elements). Two micro minerals (trace elements) are Iron and Zinc. Trace minerals are required in small amounts but are essential for important biological functions. Iron is necessary for hemoglobin formation and oxygen transport in the blood. Zinc supports immune system function, growth, wound healing, and enzyme activity. Deficiency of iron leads to anemia, weakness, and fatigue, while zinc deficiency causes reduced immunity, delayed healing, and growth retardation. Food sources include meat, eggs, whole grains, legumes, and nuts.	2	K1	CO3
9	Explain why fat-soluble vitamins can cause toxicity. Fat-soluble vitamins can cause toxicity because they are stored in the body tissues instead of being eliminated quickly. These vitamins accumulate in the liver and adipose tissue when consumed in excess. Over time, this buildup may result in harmful health effects. For example, excess Vitamin A may cause liver damage, dizziness, and birth defects, while excess Vitamin D can lead to high calcium levels causing kidney stones and organ damage. Unlike water-soluble vitamins, they are not readily excreted in urine, so supplementation must be controlled.	2	K2	CO3
10	Describe the major functions of Vitamin A. Vitamin A is essential for vision, especially in dim light, as it is required for the formation of rhodopsin in the retina. It supports immune system function by maintaining healthy skin and mucous membranes, which act as barriers against infections. Vitamin A also plays an important role in cell growth, reproduction, and tissue repair. It is needed for proper development of epithelial tissues. Deficiency can cause night blindness, xerophthalmia, and increased infection risk. Sources include carrots, liver, spinach, and dairy products.	2	K2	CO3
11	Explain the role of Vitamin D in the human body. Vitamin D plays a key role in calcium and phosphorus absorption in the intestine, which is essential for strong bones and teeth. It helps regulate blood calcium levels and supports bone mineralization. Vitamin D deficiency leads to rickets in children and osteomalacia in adults, causing weak bones and skeletal deformities. It also supports muscle function and immune health. Vitamin D can be synthesized in the skin through sunlight exposure, and dietary sources include fish liver oil, egg yolk, fortified milk, and cereals.	2	K2	CO3

12	Discuss the importance of Vitamin E. Vitamin E is an important antioxidant that protects cell membranes from oxidative damage caused by free radicals. It helps prevent lipid peroxidation, thereby maintaining the integrity of cells and tissues. Vitamin E also supports immune function and promotes healthy skin and eyes. It works synergistically with other antioxidants such as Vitamin C and selenium. Deficiency is rare but can lead to nerve and muscle damage, weakness, and vision problems. Major sources of Vitamin E include vegetable oils, sunflower seeds, almonds, peanuts, and green leafy vegetables.	2	K2	CO3
13	Explain the function of Vitamin K. Vitamin K is essential for blood clotting as it helps in the synthesis of clotting proteins such as prothrombin. Without Vitamin K, blood clot formation is impaired, leading to excessive bleeding and delayed wound healing. Vitamin K also contributes to bone health by supporting the production of proteins involved in bone mineralization. Deficiency may occur due to poor diet, liver disorders, or long-term antibiotic use. Good dietary sources include green leafy vegetables such as spinach, kale, broccoli, and cabbage, as well as fermented foods.	2	K2	CO3
14	Differentiate between fat-soluble and water-soluble vitamins. Fat-soluble vitamins dissolve in fat, require dietary fat for absorption, and are stored in the liver and adipose tissues. Excess intake may lead to toxicity. Examples include Vitamins A, D, E, and K. Water-soluble vitamins dissolve in water, are absorbed directly into the bloodstream, and are generally not stored in large quantities, so excess is excreted in urine. Examples include Vitamin C and B-complex vitamins. Water-soluble vitamins require regular intake to prevent deficiency, while fat-soluble vitamins can remain stored longer in the body.	2	K2	CO3
15	Explain the function of Thiamin (Vitamin B1). Thiamin is important for carbohydrate metabolism as it acts as a coenzyme in the conversion of glucose into energy. It is essential for proper functioning of the nervous system and helps maintain healthy muscles and heart function. Thiamin deficiency results in beriberi, which affects the cardiovascular and nervous systems. Symptoms include weakness, nerve damage, and heart failure in severe cases. It also causes Wernicke-Korsakoff syndrome in alcoholics. Thiamin is found in whole grains, legumes, pork, nuts, and fortified cereals.	2	K2	CO3
16	Describe the role of Riboflavin (Vitamin B2). Riboflavin is essential for energy production as it forms part of coenzymes such as FAD and FMN, which participate in oxidation-reduction reactions. It supports metabolism of fats, carbohydrates, and	2	K2	CO3

	proteins. Riboflavin is important for maintaining healthy skin, eyes, and mucous membranes. Deficiency causes cheilosis (cracked lips), sore throat, inflammation of the tongue, and skin disorders. Riboflavin also contributes to antioxidant activity by supporting glutathione function. Good sources include milk, eggs, meat, green vegetables, and fortified cereals.			
17	Explain the importance of Niacin (Vitamin B3). Niacin is required for the formation of NAD and NADP, which are coenzymes involved in cellular energy metabolism. It plays a key role in carbohydrate, fat, and protein metabolism. Niacin is also important for maintaining healthy skin, digestive system, and nervous system. Deficiency of niacin causes pellagra, characterized by dermatitis, diarrhea, and dementia. Severe deficiency may lead to death if untreated. Niacin is found in meat, fish, poultry, peanuts, whole grains, and legumes. It can also be synthesized from the amino acid tryptophan.	2	K2	CO3
18	Discuss the function of Vitamin B12 in the body. Vitamin B12 is essential for red blood cell formation, DNA synthesis, and normal neurological function. It supports the maintenance of the myelin sheath, which protects nerve fibers and ensures proper nerve conduction. Deficiency of Vitamin B12 leads to megaloblastic anemia, fatigue, weakness, and neurological symptoms such as numbness, memory loss, and tingling sensations. Since Vitamin B12 is mainly present in animal foods like meat, milk, eggs, and fish, vegetarians are at higher risk of deficiency unless they consume fortified foods or supplements.	2	K2	CO3
19	Explain the role of folic acid. Folic acid (Vitamin B9) plays an important role in DNA synthesis, cell division, and red blood cell production. It is essential during pregnancy because it supports fetal growth and prevents neural tube defects such as spina bifida. Folic acid is also involved in amino acid metabolism and tissue repair. Deficiency results in megaloblastic anemia, weakness, and poor growth. Pregnant women require higher folic acid intake. Major sources include green leafy vegetables, legumes, citrus fruits, and fortified cereals. Regular consumption is essential to maintain healthy blood and cell function.	2	K2	CO3
20	Describe the functions of minerals in human nutrition. Minerals are essential nutrients that help maintain body structure and regulate physiological processes. Macro minerals such as calcium and phosphorus support bone and teeth formation, while sodium and potassium maintain fluid balance and nerve conduction. Trace minerals such as iron support oxygen transport through hemoglobin, and zinc supports immunity and enzyme activity. Minerals also act as cofactors	2	K2	CO3

	for many enzymes, supporting metabolism and hormonal regulation. Deficiency of minerals may cause disorders such as anemia, osteoporosis, muscle cramps, and reduced immunity. Therefore, balanced mineral intake is necessary for good health.			
--	---	--	--	--

PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain the functions, sources, RDA and deficiency symptoms of Vitamin A.	16	K2	CO3
2	Discuss the functions, sources and deficiency disorders of Vitamin D and Vitamin K.	16	K2	CO3
3	Explain the role of water-soluble vitamins in metabolism with special reference to Thiamin and Riboflavin.	16	K2	CO3
4	Describe the classification of minerals and explain the functions and deficiency of macro and micro minerals.	16	K2	CO3
5	Apply nutritional principles to prepare a balanced diet plan rich in Vitamin C, iron and folic acid.	16	K3	CO3
6	Apply the concept of micronutrient absorption to explain the factors affecting iron and calcium absorption.	16	K3	CO3
7	Analyze the symptoms of anemia and determine the possible micronutrient deficiency with justification.	16	K4	CO3
8	Analyze the impact of fat-soluble vitamin excess and explain the health complications caused by hypervitaminosis.	16	K4	CO3

COURSE FACULTY

COURSE COORDINATOR

HOD

SCRUTINIZER



DEPARTMENT OF BIOMEDICAL ENGINEERING

QUESTION BANK

Regulation	:	2023
Course Name	:	Food & Nutrition
Course Code	:	JBM9201
Year / Semester / Branch		II/IV/ AI & ML, CS and CSE
Unit No	:	4
Unit Title	:	Dietary Guidelines and Nutritional Assessment
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1	Define Nutritional assessment. Nutritional assessment is the systematic process of evaluating the nutritional status of an individual or a population by collecting and interpreting information related to dietary intake, body measurements, biochemical tests and clinical observations. It helps in identifying malnutrition, nutrient deficiencies or excess nutrient intake.	2	K1	CO4
2	State the meaning of Rapid Rural Appraisal (RRA). Rapid Rural Appraisal is a quick and cost-effective method used to collect information about rural communities. It involves gathering data through interviews, group discussions, observation and field visits. The main purpose of RRA is to understand the living conditions, food availability, health status and socio-economic factors affecting nutrition. It is widely used in public health and development projects to assess community needs.	2	K1	CO4
3	List any two diet survey methods. Two common diet survey methods are the 24-hour dietary recall method and the food frequency questionnaire (FFQ) method. The 24-hour recall method involves asking individuals to report all foods and beverages consumed during the previous day. The FFQ method collects information on how often specific foods are consumed over a certain	2	K1	CO4

	period. These methods help estimate nutrient intake and identify dietary patterns. Diet surveys are important tools for evaluating nutrition status and planning dietary improvements in individuals or communities.			
4	Define Nutritional Anthropometry. Nutritional anthropometry refers to the measurement of the physical dimensions and composition of the human body to assess nutritional status. It includes parameters such as height, weight, body mass index (BMI), mid-upper arm circumference, skinfold thickness, and waist-to-hip ratio. These measurements help in identifying undernutrition, overweight, obesity, and growth abnormalities.	2	K1	CO4
5	Mention any two vital statistics used in indirect nutritional assessment. Two important vital statistics used in indirect nutritional assessment are infant mortality rate (IMR) and maternal mortality rate (MMR). These indicators provide information about the health and nutritional conditions of a population. A high infant mortality rate often indicates poor maternal nutrition, inadequate health care, and infections.	2	K1	CO4
6	State any two social indicators related to nutritional status. Two social indicators related to nutritional status are literacy rate and sanitation facilities. Literacy rate reflects education levels, which influence awareness about healthy food choices, hygiene, and nutrition practices. Sanitation facilities indicate environmental health conditions, as poor sanitation increases infections that can lead to malnutrition.	2	K1	CO4
7	List any two economic indicators affecting nutritional status. Two economic indicators affecting nutritional status are per capita income and employment status. Per capita income reflects the purchasing power of individuals and families, which influences their ability to buy nutritious food. Employment status affects financial stability and food security. Low income and unemployment often result in inadequate food intake, poor dietary diversity, and malnutrition..	2	K1	CO4
8	Define Primary Health Care (PHC). Primary Health Care (PHC) is the first level of health care that provides essential health services to individuals and communities. It focuses on prevention, health promotion, and basic treatment of common illnesses. PHC includes services such as immunization, maternal and child care, nutrition education, sanitation improvement, and control of infectious diseases.	2	K1	CO4
9	Explain the importance of assessing nutritional knowledge. Assessing nutritional knowledge is important because it helps identify the awareness level of individuals or communities regarding healthy eating habits, balanced diets, and nutrient requirements. Lack of nutritional knowledge often leads to poor food choices, malnutrition, obesity, and lifestyle diseases. By evaluating knowledge, health professionals can design effective nutrition education programs and awareness campaigns.	2	K2	CO4

10	Discuss the role of Rapid Rural Appraisal (RRA) in nutritional assessment. Rapid Rural Appraisal (RRA) plays an important role in nutritional assessment by providing quick information about the dietary habits, health conditions, and socio-economic factors of rural communities. It helps in identifying problems such as food insecurity, malnutrition, and lack of health facilities. RRA involves methods like group discussions, interviews, direct observation, and mapping of resources. It is cost-effective and saves time compared to detailed surveys.	2	K2	CO4
11	Explain the role of national agencies in nutrition and health development. National agencies play a crucial role in promoting nutrition and health development by formulating policies, implementing nutrition programs, and monitoring public health conditions. In India, agencies like the Ministry of Health and Family Welfare and FSSAI develop food safety regulations and nutritional guidelines. National agencies also run programs such as mid-day meal schemes, maternal nutrition programs, and immunization campaigns.	2	K2	CO4
12	Describe the contribution of international agencies in improving nutritional status. International agencies contribute significantly to improving nutritional status by providing technical support, funding, and global nutrition guidelines. Organizations like WHO, UNICEF, FAO, and World Food Programme work to reduce malnutrition and improve food security worldwide. These agencies also provide emergency food aid during disasters and support maternal and child health initiatives.	2	K2	CO4
13	Explain the 24-hour dietary recall method. The 24-hour dietary recall method is a diet survey technique used to assess an individual's food consumption over the previous 24 hours. In this method, the person is asked to recall all foods and beverages consumed, including portion sizes, preparation methods, and meal timing. This method is simple, quick, and widely used in nutritional surveys. It helps estimate nutrient intake and identify dietary patterns.	2	K2	CO4
14	Discuss the food frequency questionnaire (FFQ) method in diet surveys. The food frequency questionnaire (FFQ) method is used to evaluate usual dietary patterns over a long period, such as weeks or months. It involves asking individuals how frequently they consume specific foods, such as daily, weekly, or monthly. FFQ helps identify regular food habits and nutrient intake trends. It is commonly used in large population studies because it is easy to administer and cost-effective.	2	K2	CO4
15	Explain the significance of nutritional anthropometry in community assessment. Nutritional anthropometry is significant in community assessment because it provides measurable indicators of growth, body composition, and nutritional health. Parameters such as height,	2	K2	CO4

	weight, BMI, and mid-upper arm circumference help identify undernutrition, stunting, wasting, overweight, and obesity. Anthropometric measurements are simple, low-cost, and non-invasive, making them suitable for large-scale surveys.			
16	Explain indirect nutritional assessment using vital statistics. Indirect nutritional assessment using vital statistics involves evaluating the nutritional status of a population by analyzing health-related statistical data. Indicators such as infant mortality rate, maternal mortality rate, morbidity rate, and life expectancy provide indirect evidence of nutrition and health conditions. High mortality rates often indicate poor nutrition, low immunity, and inadequate health services. Vital statistics are useful because they reflect long-term population health trends.	2	K2	CO4
17	Describe health policy indicators used in nutritional assessment. Health policy indicators are measures used to assess the effectiveness of government health strategies and programs related to nutrition and healthcare. Examples include immunization coverage, maternal health services, availability of nutrition supplementation programs, and accessibility of primary health care. These indicators help determine whether the population is receiving adequate healthcare support to prevent malnutrition and diseases.	2	K2	CO4
18	Explain the role of social and economic indicators in nutritional status assessment. Social and economic indicators play a vital role in assessing nutritional status because nutrition is strongly influenced by living conditions, education, and income. Social indicators like literacy rate, sanitation, and access to clean water affect awareness and health. Economic indicators such as income level, poverty rate, and employment status influence food affordability and availability.	2	K2	CO4
19	Discuss the importance of coverage by primary health care in nutrition improvement. Coverage by primary health care is important in nutrition improvement because it ensures access to essential health services such as immunization, maternal and child health care, nutrition counseling, and disease prevention. PHC helps detect malnutrition early through regular health checkups and growth monitoring. It also provides supplementation programs like iron tablets, vitamin distribution, and breastfeeding guidance. Primary health care reduces infections and improves sanitation awareness, which enhances nutrient absorption.	2	K2	CO4
20	Explain the relationship between nutritional assessment and dietary guidelines. Nutritional assessment and dietary guidelines are closely related because assessment helps determine whether individuals or communities are meeting recommended dietary requirements. Dietary guidelines provide standards for balanced nutrition, while nutritional	2	K2	CO4

	assessment evaluates actual intake and health indicators.			
--	---	--	--	--

PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain the methods used for assessment of nutritional knowledge in human groups.	16	K2	CO4
2	Discuss the concept and applications of Rapid Rural Appraisal (RRA) in nutritional assessment.	16	K2	CO4
3	Describe the direct methods of nutritional assessment with emphasis on diet survey and nutritional anthropometry.	16	K2	CO4
4	Explain the indirect methods of nutritional status assessment using vital statistics, social and economic indicators.	16	K2	CO4
5	Apply appropriate diet survey methods to assess the dietary intake of a selected community group.	16	K3	CO4
6	Apply anthropometric measurements to evaluate the nutritional status of school children and interpret the findings.	16	K3	CO4
7	Analyze the relationship between primary health care coverage and nutritional status of rural populations.	16	K4	CO4
8	Analyze the impact of socio-economic indicators on malnutrition and propose suitable health policy improvements.	16	K4	CO4

COURSE FACULTY

COURSE COORDINATOR

HOD

SCRUTINIZER



DEPARTMENT OF BIOMEDICAL ENGINEERING
QUESTION BANK

Regulation	:	2023
Course Name	:	Food & Nutrition
Course Code	:	JBM9201
Year / Semester / Branch	:	III/IV/AI & ML, CS and CSE
Unit No	:	5
Unit Title	:	Nutraceuticals and Nutritional Problem
No. of Hours	:	9
Weightage in Exam (%)	:	20%

PART A

Q.No.	Questions	Marks	BTL	CO
1	Classify the types of Functional Foods with examples. Functional foods are classified as probiotics, prebiotics, and synbiotics. Probiotics contain beneficial bacteria (e.g., curd), prebiotics are non-digestible fibers (e.g., inulin), and synbiotics combine both. These improve gut health and overall wellbeing.	2	K2	CO5
2	Define Probiotics. Give one example. Probiotics are live beneficial microorganisms that improve gut flora and digestion. They help maintain intestinal microbial balance. Example: Lactobacillus in curd.	2	K1	CO5
3	Outline the role of Prebiotics in gut health. Prebiotics are non-digestible fibers that stimulate the growth of beneficial bacteria. They improve digestion and nutrient absorption. They also enhance gut immunity and prevent infections.	2	K2	CO5
4	Define Synbiotics. Synbiotics are products that combine probiotics and prebiotics. They enhance the survival and activity of beneficial bacteria. This leads to improved digestive health.	2	K1	CO5
5	Differentiate between nutrient and non-nutrient components of food. Nutrients provide energy and are essential for growth and metabolism (e.g., proteins, vitamins). Non-nutrients do not provide energy but support health (e.g., fiber). Both are important for maintaining body functions.	2	K2	CO5
6	Compare Probiotics and Prebiotics. Probiotics are live beneficial bacteria, while prebiotics are substances that promote their growth. Probiotics directly add microbes, whereas prebiotics act as food for them. Both improve gut health.	2	K2	CO5

7	Define Protein Energy Malnutrition. It is a condition caused by inadequate intake of protein and calories. It mainly affects children and leads to growth failure. Severe forms include marasmus and kwashiorkor.	2	K1	CO5
8	Outline the symptoms of Protein Energy Malnutrition. Symptoms include severe weight loss, muscle wasting, and stunted growth. Edema and irritability may also occur. Weak immunity leads to frequent infections.	2	K2	CO5
9	Define Anemia. Anemia is a condition characterized by low hemoglobin levels in the blood. It reduces oxygen-carrying capacity. This leads to fatigue and weakness.	2	K1	CO5
10	Summarize the causes of Anemia. Common causes include iron deficiency, poor diet, and blood loss. Infections and chronic diseases also contribute. Nutritional deficiencies like vitamin B12 and folate are other causes.	2	K2	CO5
11	Define Vitamin A Deficiency. Vitamin A deficiency is a condition caused by inadequate intake or absorption of vitamin A, leading to impaired vision (especially night blindness) and weakened immune function.	2	K1	CO5
12	Outline the importance of Vitamin A in vision. Vitamin A is essential for the formation of visual pigments in the retina. It helps in night vision and prevents blindness. Its deficiency leads to night blindness and xerophthalmia.	2	K2	CO5
13	Define Iodine Deficiency Disorders. Iodine deficiency disorders are a group of conditions caused by insufficient intake of iodine, which affects the normal functioning of the thyroid gland and the production of thyroid hormones. These disorders can lead to problems such as goitre, hypothyroidism, and impaired physical and mental development.	2	K1	CO5
14	Summarize the causes of iodine deficiency disorders. Iodine deficiency disorders are mainly caused by low iodine content in soil and food, leading to inadequate dietary intake. Lack of consumption of iodized salt is a major contributing factor. Poor dietary habits and limited access to iodine-rich foods also increase the risk.	2	K2	CO5
15	Name the types of Chronic Degenerative Diseases. Types of chronic degenerative diseases include diabetes mellitus, cardiovascular diseases, cancer, and osteoporosis.	2	K1	CO5
16	Differentiate between undernutrition and overnutrition. Undernutrition results from insufficient nutrient intake, leading to deficiencies. Overnutrition is due to excessive intake, causing obesity. Both negatively affect health.	2	K2	CO5
17	List two strategies to control malnutrition Promoting balanced diet and nutrition education are key strategies. Supplementation and food fortification are also effective. Government programs play an important role.	2	K1	CO5

18	Outline the role of balanced diet in preventing malnutrition. A balanced diet provides all essential nutrients in proper proportions. It supports growth, immunity, and overall health. It prevents both undernutrition and overnutrition.	2	K2	CO5
19	Interpret the impact of Vitamin A deficiency on immunity. Vitamin A deficiency weakens the immune system. It increases susceptibility to infections, especially in children. It also delays recovery from illnesses.	2	K2	CO5
20	How does iodine deficiency lead to goitre formation? Iodine deficiency reduces the production of thyroid hormones (T3 and T4). To compensate, the pituitary gland increases TSH secretion, which stimulates the thyroid gland. This continuous stimulation causes enlargement of the thyroid gland, leading to goiter.	2	K2	CO5

PART B

Q.No.	Questions	Marks	BTL	CO
1	Explain the classification of Functional Foods into Probiotics, Prebiotics, and Synbiotics with examples and health benefits.	16	K2	CO5
2	Apply your knowledge to identify nutrients and non-nutrients in a balanced diet. Discuss their roles in human health with suitable examples.	16	K3	CO5
3	Describe the magnitude, causes, clinical features, and prevention of Protein Energy Malnutrition in India.	16	K2	CO5
4	Analyze the causes, symptoms and preventive measures of Anemia with reference to the Indian population.	16	K4	CO5
5	Explain the causes, symptoms, and prevention of Vitamin A Deficiency.	16	K2	CO5
6	Apply your knowledge to explain iodine deficiency disorders, their symptoms, and suggest preventive strategies.	16	K3	CO5
7	Describe overnutrition and its impact on health with reference to Chronic Degenerative Diseases.	16	K2	CO5
8	Categorize various strategies for the control of malnutrition in India.	16	K4	CO5

COURSE FACULTY

COURSE COORDINATOR

HOD

SCRUTINIZER